

Howard Math 273, HW# 2,

Fall 2026; Instructor: Sam Hopkins; Due: Friday, October 2nd

- Let $od(n)$ denote the number of partitions of n into distinct odd parts. Show that
$$\sum_{n \geq 0} od(n)x^n = \prod_{k \geq 1} (1 + x^{2k-1}).$$
 - Let $ee(n)$ denote the number of partitions of n with an even number of even parts, and $oe(n)$ denote the number of partitions of n with an odd number of even parts. Show that
$$\sum_{n \geq 0} (ee(n) - oe(n))x^n = \prod_{k \geq 1} \frac{1}{1 + (-x)^k}.$$
 - Prove that $od(n) = ee(n) - oe(n)$ for all $n \geq 0$. **Hint:** use parts (a) and (b).
- Prove that, for any $n \geq 0$,

$$\sum_{k=0}^n \binom{2k}{k} \binom{2(n-k)}{n-k} = 4^n.$$

Hint: We discussed the generating function $\sum_{n \geq 0} \binom{2n}{n} x^n$ of the central binomial coefficients. How can you use what we proved about this generating function to deduce the desired result?

- Let $ODD(n)$ denote the subset of permutations in the symmetric group S_n that have no cycles of even size. Prove that, for any $n \geq 1$,

$$\sum_{\sigma \in ODD(n)} 2^{\#cycles(\sigma)} = 2 \cdot n!.$$

Hint: Recall that we showed

$$\sum_{n \geq 0} \left(\sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} \dots t_n^{c_n(\sigma)} \right) \frac{x^n}{n!} = e^{t_1 \frac{x}{1} + t_2 \frac{x^2}{2} + t_3 \frac{x^3}{3} + t_4 \frac{x^4}{4} + \dots},$$

where $c_k(\sigma)$ is the number of cycles of σ of size k . How can you use this generating function identity to deduce the desired result?

- Recall that a permutation $\sigma \in S_n$ is called a *derangement* if it has no fixed points, i.e., $\sigma(i) \neq i$ for all $i = 1, \dots, n$. Meanwhile, $\sigma \in S_n$ is called an *involution* if it is its own inverse, i.e., σ^2 is the identity permutation. Give a simple product formula for the number of permutations in the symmetric group S_n that are simultaneously derangements and involutions. Explain why your formula is correct.